MIL-P-60633A (MU)
3 March 1969
SUPERSEDING
MIL-P-60633 (MU)
15 December 1966

MILITARY SPECIFICATION

PROJECTILE, 40MM, XM577E1

1. SCOPE

1.1 This specification is the applicable detail specification for Projectile, 40MM, XM577E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

DRAWINGS

Munitions Command
C10542397 - Projectile, 40mm, XM577E1
IEL-10542397 - Index of Inspection Equipment Lists for Projectile, 40mm, XM577E1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Plastic defects.- All components and assemblies shall be free from blisters, bubbles, cracks, crazing, mottle, porosity or any other defects.

3.3 Sabot base integrity.- The base of the sabot shall show no evidence of failure or permanent deformation when the applicable force is applied in an axial direction.

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.4.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

* 4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

10 projectile assemblies, complete
10 sets of subassemblies
2 sets of component parts unassembled representing each mold cavity

The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Lot formation.- A lot shall consist of projectiles produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

* 4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Tool control dimensions.- Dimensions marked tool control shall be gaged at the beginning of each eight hours production from each machine in operation and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

* 4.3.2.2 Cap, Drawing B10542345, covering a detail of Drawing C10542397

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None Defined | |
| MAJOR | |
| 101. Material defective | Visual |
| MINOR | |
| 201. Hole missing | Visual |
| 202. Foreign matter | Visual |
| 203. Marking missing, incorrect or illegible | Visual |

4.3.2.3 Cup, pellet, Drawing B10542360, covering a detail of Drawing C10542397

CRITICAL

None Defined



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MAJOR** | |
| 101. Diameter of flange | Gage |
| 102. Diameter of counterbore | Gage |
| 103. Thickness of flange | Gage |
| 104. Material defective | Visual |
| 105. Flash on flange | Visual |
| **MINOR** | |
| 201. Diameter, front of cavity | Gage |
| 202. Diameter, rear of cavity | Gage |
| 203. Diameter, rear of flange (outside) | Gage |
| 204. Diameter at rear (outside) | Gage |
| 205. Depth from front face to bottom of cavity | Gage |
| 206. Length, total | Gage |
| 207. Radii of chamfers missing or incorrect | Visual (See 6.1) |
| 208. Dimples missing or incorrect | Visual (See 6.1) |
| 209. Foreign matter | Visual |

4.3.2.4 Sabot, Drawing C10542394 covering a detail of Drawing C10542397

| | |
|---|---|
| **CRITICAL** | |
| None Defined | |
| **MAJOR** | |
| 101. Diameter of bourrelet | Gage |
| 102. Diameter rear of bourrelet | Gage |
| 103. Depth of center cavity | Gage |
| 104. Drag slots missing | Visual |
| 105. Flange missing | Visual |
| 106. Stop pad missing | Visual |
| 107. Material defective | Visual |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.4 (Cont'd)

| | Defect | Method of Inspection |
|---|---|---|
| MINOR | | |
| 201. | Diameter at front of center cavity | Gage |
| 202. | Diameter at rear of center cavity | Gage |
| 203. | Diameter of obturator boss | Gage |
| 204. | Length from front face to rear of obturator boss | Gage |
| 205. | Length of stop pad | Gage |
| 206. | Length of obturator boss | Gage |
| 207. | Vent hole missing (if applicable) | Visual |
| 208. | Marking missing, incorrect or illegible | Visual |
| 209. | Excessive flash | Visual |
| 210. | Foreign matter | Visual |

4.3.2.5 Obturator, Drawing C10542355, covering a detail of Drawing C10542397.

CRITICAL

None defined

| | | |
|---|---|---|
| MAJOR | | |
| 101. | Diameter of hole | Gage |
| 102. | Diameter at large end (outside) | Gage |
| 103. | Thickness at base | Gage |
| 104. | Material defective | Visual |
| MINOR | | |
| 201. | Length, total | Gage |
| 202. | Excessive flash | Visual (see 6.1) |
| 203. | Radii or chamfers missing or incorrect | Visual (see 6.1) |
| 204. | Burr | Visual |
| 205. | Foreign matter | Visual |

* 4.3.2.6 Pellet cup assembly, Drawing C10542396 covering a detail of Drawing C10542397.

CRITICAL

None defined

| | | |
|---|---|---|
| MAJOR | | |
| 101. | Weight | Approved Scale |
| 102. | Improper assembly | Visual & Manual |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MINOR | |
| 201. Hole not sealed | Visual |
| 202. Rubber wad missing | Visual |

4.3.2.7 Projectile, 40mm, XM577E1 Drawing C10542397.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| 101. Diameter of bourrelet, max. | Gage |
| 102. Length, total, max. | Gage |
| 103. Insecure assembly | Visual & Manual |
| MINOR | |
| 201. Heat stake missing | Visual |

4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Cap.

4.3.3.1.1 Push test.- 100 caps from each lot of caps representing each mold cavity shall be subjected to this test, to determine compliance with the applicable drawing. If three or more caps completely separate prior to application of the minimum force or fail to completely separate at the maximum force, the lot represented by the sample shall be rejected. This is a destructive test. Caps so tested shall not be returned to the lot.

4.3.3.2 Sabot base integrity.- Five sabots shall be selected from each eight (8) hours production and shall be subjected to this test to determine compliance with 3.3. Failure of any sabot to comply will be cause for rejection of the lot. This is a destructive test. Parts so tested shall not be returned to the lot.

4.3.4 Inspection equipment.- Index of Inspection Equipment List No. IEL-10542397 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Cap.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.4.1.1 Push test.- The samples supplied in accordance with 4.3.3.1.1 shall be mounted on an approved device. The force will be applied to the cap until the minimum required force is reached. If complete separation has not occurred, the application of the force will be continued until complete separation has occurred or the maximum required force has been attained.

4.4.2 Sabot base integrity.- The sample supplied in accordance with 4.3.3.2 shall be mounted in an approved device in such a manner that the sabot will be held stationary. A 2000 pound force will be applied to the base of the center cavity. After the force has been released, the sabot will be visually examined, to determine compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A.- Not applicable

5.1.2 Level B.- Not applicable

5.1.3 Level C.- Packing and marking shall be in accordance with MIL-P-10025.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.- Procurement documents should specify the title, number and date of this specification.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1310 - A078



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.